Toric varieties

Shing Tak Lam

Lent 2023*

Contents

1 What is a toric variety? 1

2 Characters, cocharacters and lattices 4

3 Cocharacters and limit points - Part I 6

4 Affine toric varieties and cones 6
- 4.1 Dual cone . . . 7
- 4.2 Commutative semigroup . . . 8
- 4.3 Coordinate ring . . . 9

5 Opens in U_σ 11

6 Fans and toric varieties 11

7 Affine embeddings and toric ideals 13

Lecture 1

In this course, we do everything over $\mathbb{C}$. In particular, $\times$ means fibre product over $\mathrm{Spec}(\mathbb{C})$.

1 What is a toric variety?

Definition 1.1 (affine variety)
An *affine variety* is $V = \mathrm{Spec}(R)$, where R is a finitely generated integral[a] $\mathbb{C}$-algebra.

[a] R is an integral domain.

Fact: Given such an R, we can choose generators and relations, and write

$$R = \frac{\mathbb{C}[x_1, \ldots, x_n]}{I}$$

where I is a prime ideal. This gives an affine embedding. i.e. a closed embedding

$$V = \mathbb{V}(I) = \mathbb{A}^n$$

Note that the affine embedding is not unique.

Definition 1.2 ((complex) variety)
A *(complex) variety* is a scheme X which is integral (reduced and irreducible), separated, and of finite type.

We can avoid schemes, and glue varieties out of affine varieties. We need

*Based on lectures by Renata Picciotto. Last updated February 13, 2024.

1. a finite collection $\{V_\alpha\}_{\alpha\in A}$ of affine varieties, say $V_\alpha = \mathrm{Spec}(R_\alpha)$,
2. for $\alpha, \beta \in A$, we need Zariski opens $V_{\alpha\beta} \subseteq V_\alpha$,
3. algebraic transition maps $g_{\alpha\beta} : V_{\alpha\beta} \to V_{\beta\alpha}$, satisfying the usual cocycle conditions,
4. check that the resulting
$$X = \frac{\bigsqcup_\alpha V_\alpha}{\sim}$$
is irreducible and separated (Hausdorff).

Definition 1.3 (torus)
An *(algebraic complex) torus of dimension n* is
$$\mathrm{T}^n = (\mathbb{C}^*)^n = \mathrm{Spec}(\mathbb{C}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}]) = (\mathbb{A}^1 \setminus \{0\})^n$$

Note that this is not the same as the usual torus $(S^1)^n$. In particular, we have an affine embedding
$$\mathrm{T}^n = \mathrm{Spec}\left(\frac{\mathbb{C}[z_0, \ldots, z_n]}{\langle z_0 \cdots z_n - 1\rangle}\right) = \mathbb{V}(z_0 \cdots z_n - 1) \subseteq \mathbb{A}^{n+1}$$

Exercise: Check that we have an isomorphism of rings
$$\mathbb{C}[z_1^{\pm 1}, \ldots, z_n^{\pm 1}] \cong \frac{\mathbb{C}[z_0, \ldots, z_n]}{\langle z_0 \cdots z_n - 1\rangle}$$

The torus is also a *commutative group scheme*, that is, we have algebraic morphisms
$$\begin{aligned} e &: \bullet \to \mathrm{T}^n \\ m &: \mathrm{T}^n \times \mathrm{T}^n \to \mathrm{T}^n \\ i &: \mathrm{T}^n \to \mathrm{T}^n \end{aligned}$$
making T^n into an abelian group with identity e, multiplication m and inverse i. On the level of coordinate rings, we have the *identity section*,
$$\begin{aligned} e^\sharp : \mathbb{C}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}] &\to \mathbb{C} \\ t_i^{\pm 1} &\mapsto 1 \end{aligned}$$
The multiplication map is given by pointwise operations, i.e.
$$m((x_1, \ldots, x_n), (y_1, \ldots, y_n)) = (x_1 y_1, \ldots, x_n y_n)$$
With this, we have *comultiplication*
$$\begin{aligned} m^\sharp : \mathbb{C}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}] &\to \mathbb{C}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}] \otimes \mathbb{C}[y_1^{\pm 1}, \ldots, y_n^{\pm 1}] \\ t_i &\mapsto x_i \otimes y_i \end{aligned}$$
and the inverse map is given by
$$\begin{aligned} i^\sharp : \mathbb{C}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}] &\to \mathbb{C}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}] \\ t_i &\mapsto t_i^{-1} \end{aligned}$$

Definition 1.4 (toric variety)
A *toric variety* is a variety X with a dense open $\mathrm{T}^n \subseteq X$, and an action (i.e. algebraic map)
$$a : \mathrm{T}^n \times X \to X$$
which extends the multiplication on X.

Example 1.5

T^n is itself a toric variety.

Cohomology

Example 1.6

$\mathbb{A}^n = \mathrm{Spec}(\mathbb{C}[x_1, \ldots, x_n])$, with the torus

$$\mathrm{Spec}(\mathbb{C}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}])$$

with

$$a((t_1, \ldots, t_n), (x_1, \ldots, x_n)) = (t_1 x_1, \ldots, t_n x_n)$$

Example 1.7

Projective space

$$\mathbb{P}^n = \mathrm{Proj}(\mathbb{C}[z_0, \ldots, z_n]) = \{(z_0 : \cdots : z_n) \mid (z_0, \ldots, z_n) \neq (0, \ldots, 0)\}$$

is also a toric variety, with torus

$$\mathrm{T}^n = \{(z_0 : \cdots : z_n) \mid z_0 \cdots z_n \neq 0\}$$

with action given coordinate-wise again, i.e.

$$a((t_0 : \cdots : t_n), (z_0 : \cdots : z_n)) = (t_0 z_0 : \cdots : t_n z_n)$$

in non-homogeneous coordinates (i.e. in the affine $z_0 \neq 0$) of the torus, we can define the action by

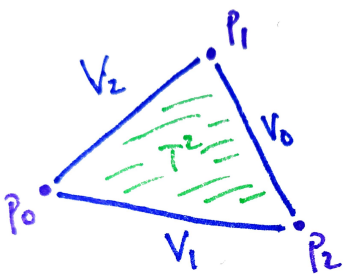
$$(t_1, \ldots, t_n) \cdot (z_0 : \cdots : z_n) = (z_0 : t_1 z_1 : \cdots : t_n z_n)$$

We can check that T^n is contained in all of the standard opens $U_i = D(z_i)$, and if we choose the usual affine coordinates, then T^n embeds into affine space as above.

Toric varieties are determined by the dimension n of the torus, and the *toric boundary* $X \setminus \mathrm{T}^n$. For example, $\mathbb{P}^2 \setminus \mathrm{T}^2$ has three components,

1. $V_0 = \mathbb{V}(z_0)$,
2. $V_1 = \mathbb{V}(z_1)$,
3. $V_2 = \mathbb{V}(z_2)$

The V_i intersect pairwise, but not all three at the same time. Each V_i is invariant under the action of the torus, so $\mathbb{P}^2$ can be represented as



In particular, this gives us a stratification of $\mathbb{P}^2$, with strata T^2, $\mathbb{C}^* = T^1$ corresponding to V_i with the vertices removed, and the three vertices.

2 Characters, cocharacters and lattices

Recall from the last lecture that $\mathrm{T}^n = (\mathbb{C}^*)^n$ is an algebraic group.

Definition 2.1 (character)
A *character* χ of T^n is a morphism $\chi : \mathrm{T}^n \to \mathbb{C}^*$ of algebraic groups. That is, $\chi : \mathrm{T}^n \to \mathbb{C}^*$ is a morphisms of schemes, such that

$$\chi(e) = 1 \quad \text{and} \quad (\chi^\sharp \otimes \chi^\sharp) \circ m^\sharp = m^\sharp \circ \chi^\sharp$$

We will write $\mathrm{Hom}_{\text{alg-gp}}(\cdot, \cdot)$ for the set of morphisms of algebraic groups.

Theorem 2.2. The characters of T^n are

$$M = \mathrm{Hom}_{\text{alg-gp}}(\mathrm{T}^n, \mathbb{C}^*) = \mathbb{Z}^n$$

Proof. First of all, note that

$$\mathrm{Hom}_{\text{alg-gp}}(\mathrm{T}^n, \mathbb{C}^*) = (\mathrm{Hom}(\mathbb{C}^*, \mathbb{C}^*))^n$$

Thus, without loss of generality $n = 1$. Now we need a $\mathbb{C}$-algebra map

$$\chi^\sharp : \mathbb{C}[x^{\pm 1}] \to \mathbb{C}[y^{\pm 1}]$$

This is determined by $\chi^\sharp(x) = p(y, y^{-1})$. Note that x is a unit in $\mathbb{C}[x^{\pm 1}]$, and so $\chi^\sharp(x)$ is a unit. Thus, $p(y, y^{-1}) = ay^k$, for some $a \in \mathbb{C}, k \in \mathbb{Z}$.

Next, we need that $e^\sharp \circ \chi^\sharp = \chi^\sharp \circ e^\sharp$. That is,

$$a = e^\sharp(\chi^\sharp(x)) = \chi^\sharp(e^\sharp(x)) = \chi^\sharp(1) = 1$$

Hence $\chi^\sharp(x) = x^k$, for some $k \in \mathbb{Z}$. Note that $\chi^\sharp(x \otimes x) = \chi^\sharp(x) \otimes \chi^\sharp(x)$ automatically, so we are done. □

The characters of T^n are of the form

$$\chi(t_1, \ldots, t_n) = t_1^{a_1} \cdots t_n^{a_n}$$

for some $(a_1, \ldots, a_n) \in \mathbb{Z}^n$. Expressions of the form

$$t_1^{a_1} \cdots t_n^{a_n}$$

are called *Laurent monomials*. We will write

$$\chi^{\mathbf{a}}(\mathbf{t}) = \mathbf{t}^{\mathbf{a}} = t_1^{a_1} \cdots t_n^{a_n}$$

for this character. The correspondence is then

$$\begin{aligned} \mathrm{Hom}_{\text{alg-gp}}(\mathrm{T}^n, \mathbb{C}^*) &\to \mathbb{Z}^n \\ \chi^{\mathbf{a}} &\mapsto \mathbf{a} \end{aligned}$$

The characters form a group. Given characters $\chi, \chi' : \mathrm{T}^n \to \mathbb{C}^*$,

$$\chi \cdot \chi'(\mathbf{t}) = \chi(\mathbf{t}) \cdot \chi'(\mathbf{t})$$

Thus, we have taht

$$\chi^{\mathbf{a}} \cdot \chi^{\mathbf{a}'} = \chi^{\mathbf{a}+\mathbf{a}'}$$

Hence the identification $\mathrm{Hom}_{\text{alg-gp}}(\mathrm{T}^n, \mathbb{C}^*) \cong \mathbb{Z}^n$ is also a group isomorphism.

Definition 2.3 (cochacter)
A *cocharacter* of T^n is

$$\lambda \in \mathrm{Hom}_{\text{alg-gp}}(\mathbb{C}^*, \mathsf{T}^n)$$

We also call them *one-parameter subgroup*.

We can check that

$$N = \mathrm{Hom}_{\text{alg-gp}}(\mathbb{C}^*, \mathsf{T}^n) \cong \mathbb{Z}^n$$

by showing that any $\lambda \in N$ is of the form

$$\lambda(t) = (t^{u_1}, \dots, t^{u_n})$$

for $\mathbf{u} = (u_1, \dots, u_n) \in \mathbb{Z}$. So we can write the above cocharacter as $\lambda_{\mathbf{u}}$ as before. N is a group, and the identification is a group isomorphism. That is,

$$\lambda_{\mathbf{u}} \cdot \lambda_{\mathbf{u}'} = \lambda_{\mathbf{u}+\mathbf{u}'}$$

Example 2.4
For $n = 2$, we can identify

$$\mathsf{T}^2 = \left\{ \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} \;\middle|\; t_1 t_2 \neq 0 \right\}$$

Under the identification $M \cong N \cong \mathbb{Z}^2$,

$$\chi^{(a,b)} \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} = t_1^a \cdot t_2^b$$

and

$$\lambda_{(u,v)}(t) = \begin{pmatrix} t^u & 0 \\ 0 & t^v \end{pmatrix}$$

Definition 2.5 (lattice)
A *lattice* is a free abelian group of finite rank.

Note that with this definition, any lattice is isomorphic to $\mathbb{Z}^n$ when we choose a basis.

Example 2.6
$\mathbb{Z}, 2\mathbb{Z}, \mathbb{Z} \oplus \mathbb{Z}, \mathbb{Z} \oplus 3\mathbb{Z}, \dots$ are all lattices.

There is a *perfect pairing*[1], given by

$$\begin{aligned} \langle \cdot, \cdot \rangle : M \times N &\to \mathbb{Z} \\ \langle \mathbf{a}, \mathbf{u} \rangle = \mathbf{a} \cdot \mathbf{u} &= a_1 u_1 + \cdots + a_n u_n \end{aligned}$$

In a coordinate-free manner,

$$\begin{aligned} M \times N = \mathrm{Hom}(\mathbb{C}^*, \mathsf{T}^n) \times \mathrm{Hom}(\mathsf{T}^n, \mathbb{C}^*) &\to \mathrm{Hom}(\mathbb{C}^*, \mathbb{C}^*) \\ (\lambda, \chi) &\mapsto \chi \circ \lambda \end{aligned}$$

Using the identification $\mathrm{Hom}(\mathbb{C}^*, \mathbb{C}^*) \cong \mathbb{Z}$ as before gives the pairing.

We can describe a torus T_N starting from a lattice N, through

$$\mathsf{T}_N = N \otimes_{\mathbb{Z}} \mathbb{C}^*$$

[1] That is, it induces isomorphism $\mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Z}) = M^\vee \cong N$ and vice versa.

with this,

$$\mathrm{Hom}(\mathbb{C}^*, \mathrm{T}_N) = N$$

Formally,

$$\mathrm{T}_N = \mathrm{Spec}(\mathbb{Z}[N] \otimes_{\mathbb{Z}} \mathbb{C}[x, x^{-1}])$$

With this, give a torus T, we get two dual lattices N, M. Conversely, given a lattice N, we get a torus T_N and a dual lattice M. Choosing a basis on one gives a basis on the other two.

3 Cocharacters and limit points - Part I

Back to toric varieties. Let X be a toric variety, with torus T. We can think of X as a "partial compactification" of T. That is, we can construct X by adding limit points[2] to T.

Given a one-parameter subgroup $\mathbf{u} \in N$, we have

$$\lambda_{\mathbf{u}} : \mathbb{C}^* \to \mathrm{T} \subseteq X$$

We can ask when does

$$\lim_{t \to 0} \lambda_{\mathbf{u}}(t)$$

exist in X. The idea will be that we can determine X by saying which $\lambda_{\mathbf{u}}$ have a limit. See later.

Example 3.1

Consider $\mathrm{T}^2 \subseteq \mathbb{P}^2_{z_0:z_1:z_2}$. The embedding is given by $(t_1, t_2) \mapsto (1 : t_1 : t_2)$. Choose $\mathbf{u} \in \mathbb{Z}^2$, so $\mathbf{u} = (u_1, u_2)$.

$$\lim_{t \to 0} \lambda_{\mathbf{u}}(t) = \lim_{t \to 0} (1 : t^{u_1} : t^{u_2})$$

All these limit points exist, since $\mathbb{P}^2$ is complete[a]. In particular, if $u_1, u_2 > 0$, the limit is $(1 : 0 : 0)$. If $u_1 < 0, u_2 > u_1$, the limit is $(0 : 1 : 0)$. Similarly if $u_2 < 0, u_1 > u_2$ then the limit is $(0 : 0 : 1)$.

The upshot is that we have partitioned $N_{\mathbb{R}} = N \otimes \mathbb{R} = \mathbb{R}^2$ into cones, so that if $\mathbf{u}_1, \mathbf{u}_2$ are in the same cone if and only if they have the same limit. In fact, we can construct $\mathbb{P}^2$ from this.

[a] Or in algebraic geometry, it satisfies the valuative criterion for properness for some choice of DVR.

Lecture 3

4 Affine toric varieties and cones

Definition 4.1 (cone)

A *(rational polyhedral) cone* is

$$\sigma \subseteq N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R} = \mathbb{R}^n$$

where

$$\sigma = \sum_{i \in I} \lambda_i u_i \mid \lambda_i \geq 0, u_1, \ldots, u_n \in \mathbb{N}$$

We will also write

$$\sigma = \mathrm{Cone}(u_1, \ldots, u_r)$$

and the set $\{u_i\}_{i=1}^r$ is the set of generators of the cone.

Example 4.2

If $N_{\mathbb{R}} \cong \mathbb{R}e$, then we have the cone $\sigma_1 = \mathrm{Cone}(e)$,
Next, $\sigma_2 = \mathrm{Cone}((1, 0))$ is
and $\sigma_3 = \mathrm{Cone}((1, 0), (0, 1))$ is

[2] In the Euclidean topology.

and $\sigma_4 = \text{Cone}((1,0),(1,2))$ is

Definition 4.3 (dimension)
The *dimension* of a cone σ is $\dim(\sigma) = \dim(V)$, where $V = \text{span}\{\sigma\} \subseteq N_{\mathbb{R}}$. A cone is *top-dimensional* if $V = N_{\mathbb{R}}$.

Definition 4.4 (strongly convex)
σ is *strongly convex* if it does not contain any lines through the origin. Equivalently, $\sigma \cap -\sigma = \{0\}$.

We would like to associate a to a cone σ, an affine toric variety $U_\sigma = \text{Spec}(R_\sigma)$. That is, we would like to define

$$R_d \subseteq \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$$

a sub-$\mathbb{C}$-algebra.

We will do this as follows:

1. We will associate a *dual cone* $\sigma^\vee \subseteq M_{\mathbb{R}}$,
2. Then we will generate a commutative semigroup S_σ,
3. from this, we will define R_σ.

4.1 Dual cone

Definition 4.5 (dual cone)
The *dual cone* $\sigma^\vee$ of σ is

$$\check{\sigma} = \{m \in M_{\mathbb{R}} \mid \langle u, m\rangle \geq 0 \text{ for all } u \in \sigma\} \subseteq M_{\mathbb{R}}$$

where we extend the pairing to $N_{\mathbb{R}} \otimes M_{\mathbb{R}} \to \mathbb{R}$.

Example 4.6
Using the same examples as before, we have the dual cones $\check{\sigma_1}$
and $\check{\sigma_2}$,
Note here we see that the dual cone depends on the embedding (or lattice). Moroever, the dual of a strongly convex cone does not have to be strongly convex.

Fact: $\check{\check{\sigma}} = \sigma$.

Lemma 4.7. The dual $\check{\sigma}$ of a cone σ is a cone.

Proof. We'll define an algorithm to find $m_1, \dots, m_s \in M$, such that

$$\check{\sigma} = \text{Cone}(m_1, \dots, m_s)$$

From linear algebra, for $m \in M = N^\vee$, we have a hypersurface

$$m^\perp = H_m = \{u \in N_{\mathbb{R}} \mid \langle u, m\rangle = 0\}$$

We also have half spaces

$$H_m^+ = \{u \in N_{\mathbb{R}} \mid \langle m, u\rangle \geq 0\}$$

Definition 4.8 (supporting hyperplane)
A *supporting hyperplane* of σ is H_m, such that $\sigma \subseteq H_m^+$. Equivalently, $m \in \check{\sigma}$.

Now note that

$$\check{\sigma} = \text{Cone}(m_1, \dots, m_s) \iff \sigma = \bigcap_{i=1}^{s} H_{m_i}^+$$

So to construct $\check{\sigma}$, we need to express σ as an intersection of half spaces.

For example, $\check{\sigma}_3 = \text{Cone}((1,0),(0,1))$

and similarly, $\check{\sigma}_4 = \text{Cone}((2,-1),(0,1))$.

In general, we can do this by identifying the *facets* $\tau \preccurlyeq \sigma$, and taking the corresponding dual element $m_\tau \in M$, and

$$\check{\sigma} = \text{Cone}(m_\tau \mid \tau \preccurlyeq \sigma)$$

when σ is top dimensional.

Definition 4.9
A *face* $\tau \preccurlyeq \sigma$ is the intersection $\sigma \cap H_m$, for a supporting hyperplane H_m. Note that the faces are all cones, and we call τ a *facet* if $\dim(\tau) = \dim(\sigma) - 1$, and τ an *edge* if $\dim(\tau) = 1$.

Note by taking $m = 0$, $\sigma \preccurlyeq \sigma$.

When σ is not top dimensional, then $\text{span}_{\mathbb{R}}(\sigma) = W \neq N_{\mathbb{R}}$. Write $\overline{\sigma}$ for σ considered as a subspace of W. Now $\overline{\sigma}$ is top-dimensional, and so we can the dual

$$\check{\overline{\sigma}} \subseteq \frac{M_{\mathbb{R}}}{W^\perp}$$

Say

$$\check{\overline{\sigma}} = \text{Cone}(\overline{m_1}, \dots, \overline{m_s})$$

where

$$\overline{m_i} \in \frac{M_{\mathbb{R}}}{W^\perp}$$

Choose representatives $m_i \in M_{\mathbb{R}}$, and choose a basis $m_{s+1}, \dots, m_k$ of $W^\perp$ in $M_{\mathbb{R}}$. With this,

$$\check{\sigma} = \text{Cone}(m_1, \dots, m_s, \pm m_{s+1}, \dots, \pm m_k)$$

□

4.2 Commutative semigroup

Definition 4.10 (commutative semigroup)
A *commutative semigroup* is $(S, +)$, such that $+$ is an associative commutative binary operation on S.

In particular, a semigroup with identity is called a *monoid*. S is *finitely generated* if there exists $A \subseteq S$ finite, such that

$$S = \left\{ \sum_n \lambda_n a_n \;\middle|\; \lambda_n \in \mathbb{Z}_{>0}, a_n \in A \right\}$$

S is *affine* if it is finitely generated, and $(S, +)$ is a subsemigroup of a lattice $(\mathbb{Z}^n, +)$.

Example 4.11
Any group is a semigroup. Given a (not necessarily unital) ring R, R under multiplication is a semigroup. For example, $\mathbb{Z}_{>0}$ is a semigroup under multiplication, but this is not finitely generated.

Lecture 4

Definition 4.12

Let σ be a cone. Define

$$S_\sigma = \check{\sigma} \cap M \subseteq M_{\mathbb{R}}$$

for the semigroup associated to σ.

This is a semigroup as cones are closed under addition. Now we want to show that this is affine. This does not follow immediately from being a subgroup of $(\mathbb{Z}^n, +)$. For example, consider

$$S = \{(x, y) \mid x \leq \sqrt{2}y\}$$

This is a semigroup, but it is not finitely generated.

Lemma 4.13 (Gordan's lemma)**.** If σ is a cone, then S_σ is a finitely generated semigroup.

Proof. We know that $\check{\sigma} = \text{Cone}(m_1, \ldots, m_r)$, for some $m_i \in M$. Say $x \in S_\sigma$. By definition,

$$x = \sum_{i=1}^{r} \alpha_i m_i$$

where $\alpha_i \in \mathbb{R}_{\geq 0}$. Write

$$\alpha_i = a_i + b_i$$

where $a_i = \lfloor \alpha_i \rfloor$, $b_i = \{\alpha_i\} \in [0, 1)$. Then

$$x = \sum_i a_i m_i + \sum_i b_i m_i$$

Now $x \in M$, and $\sum_i a_i m_i \in M$, and M is a group, so

$$\sum_i b_i m_i \in M$$

On the other hand, $\sum b_i m_i$ is contained in a compact set

$$K = \left\{ \sum c_i m_i \mid 0 \leq c_i \leq 1 \right\}$$

Since M is discrete, K is compact, $K \cap M$ is finitely many points, say $K \cap M = \{m_{r+1}, \ldots, m_\ell\}$. Then S_σ is generated by

$$m_1, \ldots, m_r, m_{r+1}, \ldots, m_\ell$$

so it is finitely generated. □

Example 4.14

When $\sigma = \text{Cone}(1)$, then $\check{\sigma} = \text{Cone}(1)$, and so S_σ is generated by 1.

Now if $\sigma = \text{Cone}(e_1) \subseteq \mathbb{R}^2$, then $\check{\sigma} = \text{Cone}(e_1, e_2, -e_2)$, and these are the generators of S_σ.

If $\sigma = \text{Cone}(e_1, e_2) \subseteq \mathbb{R}^2$, then $\check{\sigma} = \text{Cone}(e_1, e_2)$, which are the generators of S_σ.

Finally, if $\sigma = \text{Cone}(e_1, e_1 + 2e_2)$, then $\check{\sigma} = \text{Cone}(e_2, 2e_1 - e_2)$. Here, Gordan's lemma shows that S_σ is generated by

$$e_2, 2e_1 - e_2, e_1$$

4.3 Coordinate ring

Definition 4.15

The algebra $\mathbb{C}[S]$ of a semigroup $(S, +)$ is the $\mathbb{C}$-algebra of finite sums

$$\mathbb{C}[S] = \left\{ \sum_i a_i x^{s_i} \mid a_i \in \mathbb{C}, s_i \in S \right\}$$

with $x^s x^{s'} = x^{s+s'}$.

Example 4.16

The semigroup $\mathbb{Z}^n$ is generated by $e_1, \dots, e_n, -e_1, \dots, -e_n$, so

$$\mathbb{C}[\mathbb{Z}^n] = \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$$

where $x_i^{\pm 1} = x^{\pm e_i}$.

Lemma 4.17. If $(S, +)$ is a finitely generated semigroup, $\mathbb{C}[S]$ is a finitely generated $\mathbb{C}$-algebra. If $(S, +)$ is affine, then $\mathbb{C}[S]$ is the algebra of an affine variety.

Proof. If S is generated by $A = \{s_1, \dots, s_\ell\}$, then $\{x^{s_1}, \dots, x^{s_\ell}\}$ generates $\mathbb{C}[S]$ as a $\mathbb{C}$-algebra.

Now suppose S is affine. Then $\mathbb{C}[S]$ is finitely generated by the above, and we have that $(S, +) \subseteq (\mathbb{Z}^n, +)$, so we have

$$\mathbb{C}[S] \subseteq \mathbb{C}[\mathbb{Z}^n] = \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$$

So $\mathbb{C}[S]$ is a subring of an integral domain, hence it is an integral domain. To conclude, every finitely generated integral domain is the coordinate ring of an affine variety. □

Define $R_\sigma = \mathbb{C}[S_\sigma]$. From this, we know that $U_\sigma = \operatorname{Spec}(R_\sigma)$ is an affine variety.

Example 4.18

For $\sigma = \operatorname{Cone}(1)$, let $x = x^{e_1}$, then $\mathbb{C}[S_\sigma] = \mathbb{C}[x]$, and so $U_\sigma = \mathbb{A}^1$. The inclusion $\mathbb{C}[x] \subseteq \mathbb{C}[x^{\pm 1}]$ defines the torus.

When $\sigma = \operatorname{Cone}(e_1) \subseteq \mathbb{R}^2$, let $x = x^{e_1}, y = x^{e_2}$, then $\mathbb{C}[S_\sigma] = \mathbb{C}[x, y^{\pm 1}]$, and $U_\sigma = \mathbb{A}^1 \times \mathbb{C}^*$. The torus is given by the inclusion

$$\mathbb{C}[x, y^{\pm 1}] \subseteq \mathbb{C}[x^{\pm 1}, y^{\pm 1}]$$

When $\sigma = \operatorname{Cone}(e_1, e_2) \subseteq \mathbb{R}^2$, set $x = x^{e_1}, y = x^{e_2}$, then $U_\sigma = \operatorname{Spec}(\mathbb{C}[x, y]) = \mathbb{A}^2$.

Finally, when $\sigma = \operatorname{Cone}(e_1, e_1 + 2e_2)$. Let $A = e_1, B = 2e_1 - e_2, C = e_2$. Then S_σ is generated by A, B, C. We have a relation $B + A = 2C$. Define $x = x^A, y = x^B, z = x^C$, then

$$\mathbb{C}[S_\sigma] = \frac{\mathbb{C}[x, y, z]}{\langle xy = z^2 \rangle}$$

With this, $U_\sigma = \operatorname{Spec}(\mathbb{C}[S_\sigma]) = \mathbb{V}(xy - z^2) \subseteq \mathbb{A}^3$. In particular, this is a *singular* toric variety. In this case, the torus is given by

$$\frac{\mathbb{C}[x, y, z]}{\langle xy - z^2 \rangle} \subseteq \mathbb{C}[x^{\pm 1}, z^{\pm 1}] = \mathbb{C}[M]$$

To see this, we have an inclusion

$$\frac{\mathbb{C}[x, y, z]}{\langle xy - z^2 \rangle} \subseteq \frac{\mathbb{C}[x^{\pm 1}, y, z^{\pm 1}]}{\langle xy - z^2 \rangle}$$

Remark 4.19. We know that $S_\sigma \subseteq M$, and so we have an inclusion of algebras $\mathbb{C}[S_\sigma] \subseteq \mathbb{C}[M]$. A fact from commutative algebra is that, the image of the map

$$\mathrm{T}^n = \operatorname{Spec}(\mathbb{C}[M]) \to U_\sigma = \operatorname{Spec}(\mathbb{C}[S_\sigma])$$

is dense, as these are integral domains.

Remark 4.20. T^n may map to some smaller torus $\mathrm{T}^{n'} \subseteq U_\sigma$, with $n' \leq n$. We claim that if σ is strongly convex, then

$T^n \subseteq U_\sigma$ as a dense open subset. See examples sheet 1, questions 1 and 3. For this, note that σ is strongly convex if and only if $\check{\sigma}$ is top dimensional.

From now on, restrict to strongly convex σ.

What remains is to construct the torus action. Recall $T_N = \mathrm{Spec}(\mathbb{C}[M])$. We will define this by

$$\begin{aligned} a^\sharp : \mathbb{C}[S_\sigma] &\to \mathbb{C}[S_\sigma] \otimes \mathbb{C}[M] \\ \chi^{\mathbf{m}} &\mapsto \chi^{\mathbf{m}} \otimes \chi^{\mathbf{m}} \end{aligned}$$

This extends

$$\begin{aligned} m^\sharp : \mathbb{C}[M] &\to \mathbb{C}[M] \otimes \mathbb{C}[M] \\ \chi^{\mathbf{m}} &\mapsto \chi^{\mathbf{m}} \otimes \chi^{\mathbf{m}} \end{aligned}$$

Remark 4.21. The reason this works is the fact that $S_\sigma \subseteq M$ tells us which functions on U_σ are monomials.

With this all in mind, given a strongly convex cone σ, we have an affine toric variety U_σ, with torus T_N. We will see that

$$\begin{array}{ccc} \{\text{affine toric varieties}\} & \longleftrightarrow & \{\text{affine semigroups}\} \\ \uparrow & & \uparrow \\ \{\text{affine } \textit{normal} \text{ toric varieties}\} & \longleftrightarrow & \{\text{strongly convex cones}\} \end{array}$$

5 Opens in U_σ

We will see that $\tau \preccurlyeq \sigma$, correspond to principal opens $U_\tau \subseteq U_\sigma$. Recall that a face is $\tau = \sigma \cap H_m$ for some $m \in \check{\sigma} \cap M = S_\sigma$.

Theorem 5.1. If $\tau = \sigma \cap H_m$, then

$$\mathbb{C}[S_\tau] = \mathbb{C}[S_\sigma]_{\chi^{\mathbf{m}}} = \mathbb{C}[S_\sigma][\chi^{\pm \mathbf{m}}]$$

Proof. First, since $\tau \in H_m$, $\langle u, m \rangle = 0$ for all $u \in \tau$. So m and $-m$ are in $\check{\tau} \cap M = S_\tau$. Also, if $m' \in S_\sigma$, then $\langle m', u \rangle \geq 0$ for all $u \in \tau$. Putting these together,

$$\mathbb{C}[S_\sigma]_{\chi^{\mathbf{m}}} \subseteq \mathbb{C}[S_\tau]$$

For the reverse inclusion, take $m' \in S_\tau \setminus S_\sigma$. We want to show $m' + km \in S_\sigma$ for some $k \in \mathbb{Z}$. For $u \in \sigma \setminus \tau$, $\langle m', u \rangle < 0$. But we also have that $\langle m, u \rangle > 0$, so we can pick k large enough such that $\langle m' + km, u \rangle > 0$. This tells us that $m' + km \in S_\sigma$[3]. □

Example 5.2

Let $\sigma = \mathrm{Cone}(e_1, e_2)$. Let $\star = (0, 0)$. This is a face, and $S_\star = \mathbb{C}[x^{\pm 1}, y^{\pm 1}]$. Hence $U_\star = \mathrm{T} \subseteq \mathbb{A}^2$.

Now if τ_1 is the face $y = 0$, then we get

$$\mathbb{C}[S_{\tau_1}] = \mathbb{C}[x, y]_{\chi^{e_2} = y} = \mathbb{C}[x, y^{\pm 1}]$$

So

$$U_{\tau_1} = \mathbb{A}^1 \times \mathbb{C}^*$$

6 Fans and toric varieties

[3] Since we only need to show positivity on the finitely many generators of the cone.

Definition 6.1 (fan)

A *fan* Σ is a finite collection of strongly convex cones $\sigma \subseteq N_{\mathbb{R}}$, such that

1. for all $\sigma \in \Sigma$, $\tau \preccurlyeq \sigma$, then $\tau \in \Sigma$.
2. for any $\sigma_1, \sigma_2 \in \Sigma$, $\tau = \sigma_1 \cap \sigma_2$ is a face of both.

Notation 6.2. The *support* of Σ is

$$|\Sigma| = \bigcup_{\sigma \in \Sigma} \sigma \subseteq N_{\mathbb{R}}$$

and we will write

$$\Sigma(r) \subseteq \Sigma$$

for the subset of r-dimensional cones.

Example 6.3

First, let $\sigma = \mathrm{Cone}(e) \subseteq \mathbb{R}$, then $\Sigma_1 = \{\sigma, -\sigma, \star\}$ is a fan.

Next, we have seen the fan Σ for $\mathbb{P}^2$. Here, $\Sigma(0) = \{\star\}$, $\Sigma(1) = \{\mathrm{Cone}(e_1), \mathrm{Cone}(e_2), \mathrm{Cone}(-e_1 - e_2)\}$ and so on.

Next, consider the cone given by $\sigma_0 = \mathrm{Cone}(e_1, e_1 + e_2)$, $\sigma_1 = \mathrm{Cone}(e_1 + e_2, e_2)$ and their faces.

Finally, consider the rays passing through $(0,1), (1,0), (0,-1), (-1,a)$, for soem $a \in \mathbb{Z}_{>0}$.

By the above construction, we can associate to each $\sigma \in \Sigma$ an affine toric variety, and if $\tau \preccurlyeq \sigma$ is a face, then $U_\tau \subseteq U_\sigma$ is an open. Thus, we can glue the U_σs together, using the explicit open embeddings. This gives an (abstract) variety X_Σ.

Remark 6.4. The second condition will imply separatedness.

Example 6.5

For the first example, $U_\sigma = \mathrm{Spec}(\mathbb{C}[x])$, and $U_{-\sigma} = \mathrm{Spec}(\mathbb{C}[x^{-1}])$. Finally, the origin corresponds to $\mathrm{Spec}(\mathbb{C}[x, x^{-1}])$. So we get $\mathbb{A}^1$ and $\mathbb{A}^1$ glued along $\mathbb{C}^*$, which is $\mathbb{P}^1$.

Lecture 6

Example 6.6

For the fan with for $\mathbb{P}^2$ as above, the dual fan is

diagram

Let

$$x = \chi^{(1,0)}$$
$$y = \chi^{(0,1)}$$

In this case,

$$U_{\sigma_0} = \mathrm{Spec}(\mathbb{C}[x, y])$$
$$U_{\sigma_1} = \mathrm{Spec}(\mathbb{C}[x^{-1}, x^{-1}y])$$
$$U_{\sigma_2} = \mathrm{Spec}(\mathbb{C}[y^{-1}, xy^{-1}])$$

We would like to identify X_Σ with $\mathbb{P}^2$, with homogeneous coordinates $[z_0 : z_1 : z_2]$. If

$$x = \frac{z_1}{z_0} \quad \text{and} \quad y = \frac{z_2}{z_0}$$

This identifies U_{σ_0} with the open $\{z_0 \neq 0\}$. Similarly, we can identify the other opens.

Example 6.7

Consider now the fan given by

$$\sigma_1 = \text{Cone}(e_1, e_1 + e_2)$$
$$\sigma_2 = \text{Cone}(e_1 + e_2, e_2)$$

The dual cones are

$$\sigma_1^\vee = \text{Cone}(e_2, e_1 - e_2)$$
$$\sigma_2^\vee = \text{Cone}(e_1, -e_1 + e_2)$$

Note that the dual of a fan is not a fan. In this case,

$$U_{\sigma_1} = \text{Spec}(\mathbb{C}[y, xy^{-1}])$$
$$U_{\sigma_2} = \text{Spec}(\mathbb{C}[x, x^{-1}y])$$

Set $t = xy^{-1}$. Then we have a projection map to $\mathbb{P}^1$. In particular, this is the total space of $\mathcal{O}(-1)$[a], which is

$$\{xz_1 = yz_0\} \subseteq \mathbb{C}^2_{x,y} \times \mathbb{P}^1_{[z_0,z_1]}$$

Setting $t = z_0/z_1$ gives the identification. This is also the blowup of $\mathbb{A}^2$ at 0.

[a] which is the tautological bundle. Using relative spec, we can also write this as

$$\text{Spec}_{\mathbb{P}^1}\left(\text{Sym}^\bullet_{\mathbb{P}^1} \mathcal{O}(1)\right)$$

More generally, taking the spec of the symmetric algebra associates a vector bundle to a locally free sheaf. The vector bundle has a trivialising open affine cover.

Conversely, for a vector bundle, take the sheaf of sections.

Fact: If Σ_1, Σ_2 are fans, then so is $\Sigma_1 \times \Sigma_2$, and $X_{\Sigma_1 \times \Sigma_2} = X_{\Sigma_1} \times X_{\Sigma_2}$.

7 Affine embeddings and toric ideals

Lecture 7

We'll see two alternative ways of describing toric varieties:

1. toric ideals, $V = \mathbb{V}(I_L) \subseteq \mathbb{A}^s$, where I_L is a toric ideal,
2. torus embedding, $V = \overline{\Phi_{\mathscr{A}}(T)}$, where $\Phi_{\mathscr{A}} : T \to \mathbb{A}^s$ given by monomials.

Recall for $\sigma = \text{Cone}((1,0),(1,2))$, we have $\sigma^\vee = \text{Cone}((0,1),(2,-1))$, and

$$S_\sigma = \mathbb{Z}_{\geq 0}\mathscr{A}$$

where $\mathscr{A} = \{A, B, C\}$, where $A = (0,1), B = (1,0), C = (2,-1)$. We have a map

$$\pi : \mathbb{Z}^3_{A,B,C} \to M \cong \mathbb{Z}^2$$
$$A \mapsto (0,1)$$
$$B \mapsto (1,0)$$
$$C \mapsto (2,-1)$$

As a matrix,

$$\begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \end{pmatrix}$$

Then

$$\ker(\pi) = \mathbb{Z} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

If we restrict to $\mathbb{N}^3_{A,B,C}$ and S_σ, then we have a surjection of semigroups. In particular, $S_\sigma = \mathbb{N}^3_{A,B,C}/\sim$, where $\sim$ is the relation $A + C = 2B$. On algebras,

$$\mathbb{C}[S_\sigma] = \frac{\mathbb{C}[\mathbb{N}_{A,B,C}]}{I_L}$$

Let $x = \chi^A, y = \chi^B, z = \chi^C$. Then $A + C = 2B$ corresponds to $xz = y^2$. Then $I_L = \langle xz - y^2 \rangle$. In particular, this shows that we have an affine embedding $U_\sigma = \mathbb{V}(I_L) \subseteq \mathbb{A}^3$.

On the other hand, χ^A, χ^B, χ^C are maps $T \to \mathbb{C}^*$. We can define a map

$$\begin{aligned} \Phi_{\mathscr{A}} : T &\to \mathbb{A}^3 \\ (s,t) &\mapsto \left(\chi^A(s,t), \chi^B(s,t), \chi^C(s,t)\right) = (t, s, s^2t^{-1}) \end{aligned}$$

In this case,

$$U_\sigma = \overline{\Phi_{\mathscr{A}}(T)}$$

Moreover, $\Phi_{\mathscr{A}}$ with coordinates s, t, acts on U_σ by

$$(s,t)(x,y,z) = (tx, sy, s^2t^{-1}z)$$

More generally, if S is an affine semigroup, generated by $\mathscr{A} = \{m_1, \dots, m_s\}$, where $m_i \in M$, a lattice. So $S = \mathbb{Z}_{\geq 0}\mathscr{A}$. We can associate an ideal and a torus embedding as above.

Definition 7.1

A *lattice ideal* is an ideal I_L in $\mathbb{C}[x_1, \dots, x_s]$, of the form

$$I_L = \left\langle x^\alpha - x^\beta \mid \alpha - \beta \in L \right\rangle$$

for some lattice L. Note here we use the multi-index notation.

I_L is *toric* if it is prime.

Now given S as above, consider the map

$$\pi : \bigoplus_{i=1}^{s} \mathbb{Z} \cdot m_i \to M$$

and $L = \ker(\pi)$ is a sublattice. These relations give an ideal

$$I_L = \left\langle x^{\ell^+} - x^{\ell^-} \;\middle|\; \ell^+ - \ell^- \in L; \ell^+, \ell^- \in \bigoplus \mathbb{Z}_{\geq 0} \cdot m_i \right\rangle$$

This is an ideal of

$$\mathbb{C}[x_1, \dots, x_s] = \mathbb{C}\left[\bigoplus_{i=1}^{s} \mathbb{Z}_{\geq 0} \cdot m_i\right]$$

where $x_i = \chi^{m_i}$. We can define a variety $V = \mathbb{V}(I_L) \subseteq \mathbb{A}^s$.

Now for $\Phi_{\mathscr{A}}$, we recall that $M = \mathbb{Z}S$, and $T_N = \operatorname{Spec}(\mathbb{C}[M])$. We can define

$$\begin{aligned} \Phi_{\mathscr{A}} : T_N &\to \mathbb{A}^s \\ \mathbf{t} &\mapsto (\chi^{m_1}(\mathbf{t}), \dots, \chi^{m_s}(\mathbf{t})) \end{aligned}$$

Theorem 7.2. If V is an affine variety, then the following are equivalent:

(i) V is an affine toric variety,

(ii) $V = \operatorname{Spec}(\mathbb{C}[S])$ for an affine semigroup S,

(iii) $V = \mathbb{V}(I_L)$ for a toric ideal I_L,

(iv) $V = \overline{\Phi_{\mathscr{A}}(T)}$ for some finite collection $\mathscr{A}$ generating an affine semigroup.

Proof. We have seen (ii) implies (iii) and that (iii) implies (iv). To see that (iv) implies (i), note that $\Phi_{\mathscr{A}}$ gives an action of T on $\mathbb{A}^s$, and if $V = \overline{\Phi_{\mathscr{A}}(T)}$, then $\Phi_{\mathscr{A}}(T) \subseteq V$ is a dense open torus, and we just need to check that for $t \in T$, $t \cdot V \subseteq V$. Now $t \cdot V$ contains $\Phi_{\mathscr{A}}(T)$, so it is contained in the closure of $\Phi_{\mathscr{A}}(T)$, which is V.

For (iv) implies (i), we need a lemma from representation theory:

Lemma 7.3. If $A \subseteq \mathbb{C}[M]$ is a subrepresentation of T_N, then A splits into

$$A = \bigoplus_{m \in S} \mathbb{C} \cdot \chi^m$$

for some subset[a] $S \subseteq M$.

[a] Since the multiplicity in $\mathbb{C}[M]$ is one for each eigenspace, each eigenspace appears either zero or one times, so we can consider a *subset* without thinking about multiplicity.

Note that here, multiplication by t defines a map $\mathrm{T}_N \to \mathrm{T}_N$, which in turn is a map on coordinate rings $\mathbb{C}[M] \to \mathbb{C}[M]$.

Applying the lemma to $\mathbb{C}[V] \to \mathbb{C}[M]$. The collection S of monomial appearing in $\mathbb{C}[V]$ needs to form a semigroup S, as $\mathbb{C}[V]$ is a ring. This is finitely generated as $\mathbb{C}[V]$ is finitely generated, as it is the coordinate ring of an affine variety. □

Under the correspondence

$$\{\text{affine toric variety}\} \leftrightarrow \{\text{affine semigroups}\}$$

Restricting, we get

$$\{\text{normal affine toric varieties}\} \leftrightarrow \{\text{saturated semigroups}\}$$

Example 7.4

Let $M = \mathbb{Z}$, and $S = \langle 2, 3 \rangle = \{0, 2, 3, \dots\}$. But we can still construct a toric variety

$$V = \mathrm{Spec}(\mathbb{C}[S])$$

So we have a map

$$\begin{aligned} \mathbb{Z}^2 &\to \mathbb{Z} \\ e_1 &\mapsto 2 \\ e_2 &\mapsto 3 \end{aligned}$$

and the kernel is generated by $2e_2 - 3e_1$. Now take $x = \chi^{e_1}, y = \chi^{e_2} \in \mathbb{C}[\mathbb{Z}^2]$. We have an associated ideal, $I_L = \langle y^2 = x^3 \rangle$. Then $C = \mathrm{Spec}(\mathbb{C}[x, y]/I_L) = \mathbb{V}(I_L)$ is the affine toric variety associated to S. In particular, this is a cuspidal cubic.

For a parametrisation, we have

$$\begin{aligned} \Phi_{\mathscr{A}} : \mathbb{C}^* &\to \mathbb{A}^2 \\ t &\mapsto (t^2, t^3) \end{aligned}$$

and $C = \overline{\Phi_{\mathscr{A}}(\mathbb{C}^*)}$.

On the other hand, this is not a normal variety.

Lecture 8

Index

affine variety, 1

commutative group scheme, 2

toric boundary, 3
toric variety, 2
torus, 2

variety, 1